

UNIVERSITY OF PRINCE EDWARD ISLAND

Machine Learning

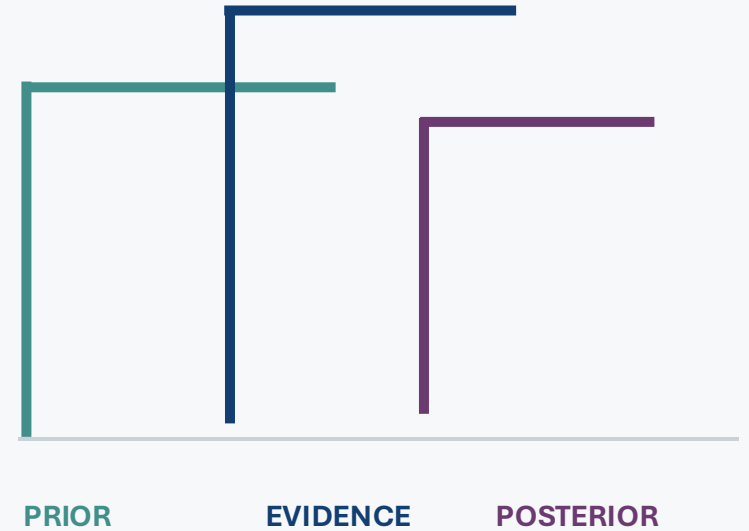
Core Concepts & Bayesian Learning

PRESENTED BY

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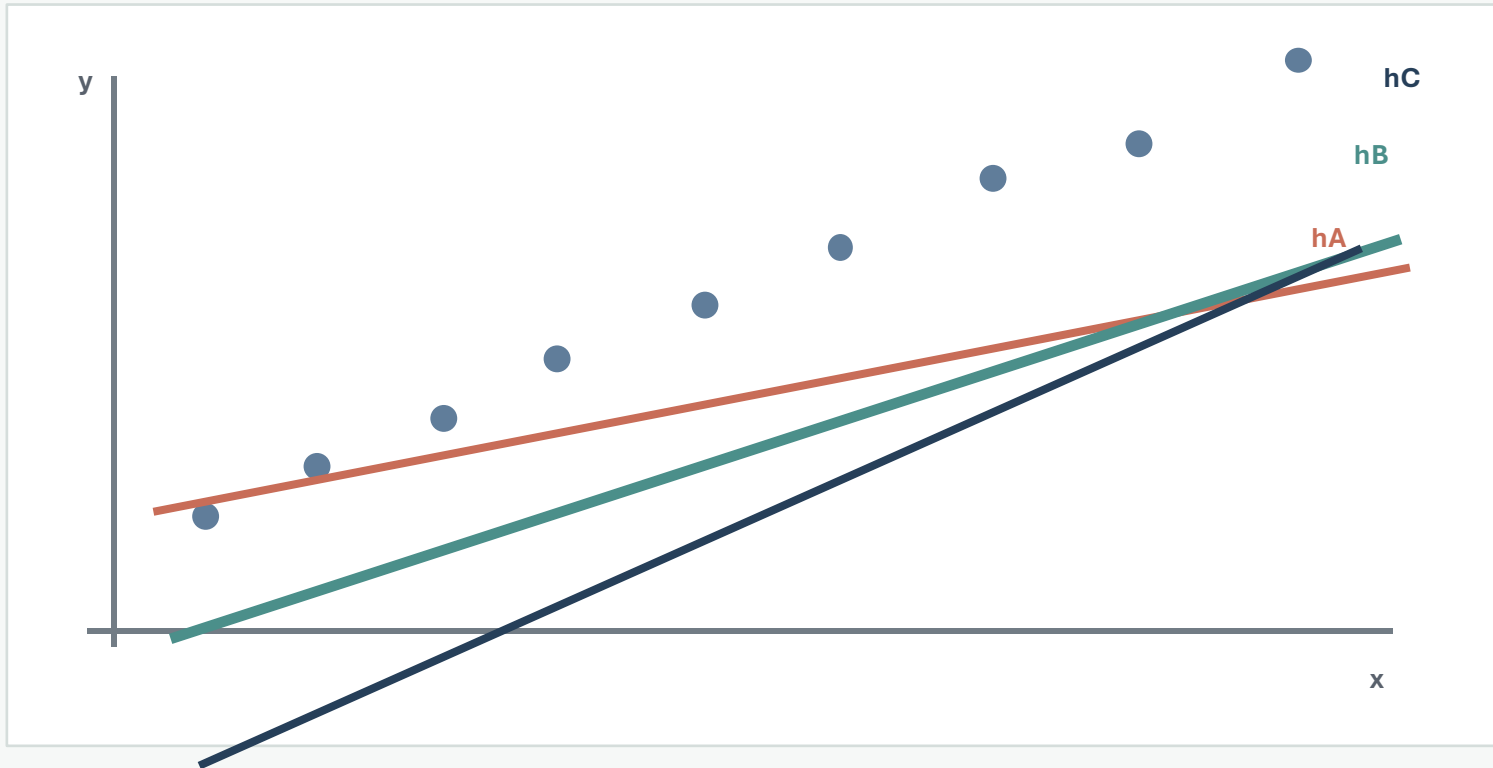
September 23, 2026

LEARNING FROM EVIDENCE



A hypothesis is one candidate explanation of the data

Change the parameters, and the line and its predictions change.



HYPOTHESIS FUNCTION

$$h\theta(x) = \theta_0 + \theta_1 x$$

θ_0 moves the line up or down

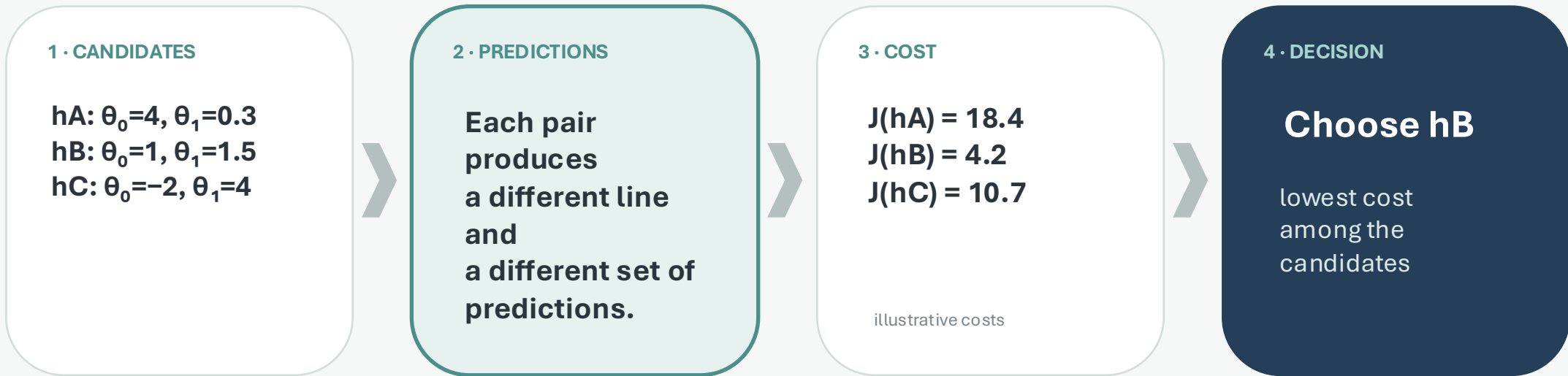
θ_1 changes the line's tilt

Every parameter pair defines a different candidate model.

Learning means selecting parameter values that make the predictions fit the evidence better.

Learning a line is already a model selection problem

We compare candidate hypotheses, measure their error, and keep the one the evidence supports.



Challenge 1 moved one level up: the polynomial degree defines the hypothesis family.

Validation evidence helps us choose among families, then stronger evidence can revise the choice.

Challenge 1: What Were We Actually Trying to Decide?



The goal was not to discover “the correct polynomial degree.”

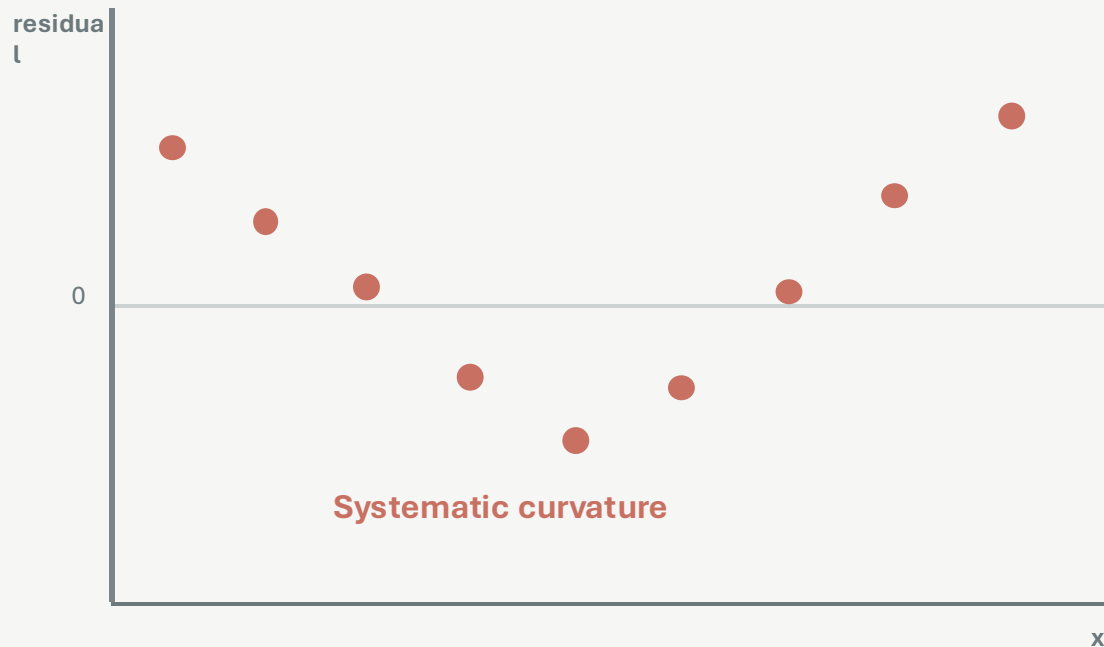
The goal was to decide which model the available evidence supports.

When is the evidence strong enough to choose a model?

The residual plot showed the line was missing curvature

FROM THE NOTEBOOK

CHALLENGE 1 · LINEAR BASELINE RESIDUALS



The notebook output showed residuals positive at the edges and negative near the center.

ASK FIRST

What would we expect if the linear model captured the relationship well?

Residuals should be scattered around zero without a systematic pattern.

Pattern matters more than being “very close to zero.”

So Which Degree Wins?

POLYNOMIAL DEGREE	VALIDATION RMSE
Degree 2	3.340
Degree 3	3.333
Degree 4	3.337

THE SMALLEST NUMBER

Degree 3

appears to win

Degree 3 compared with Degree 2

0.007 RMSE

Degree 3 wins. Are we done?

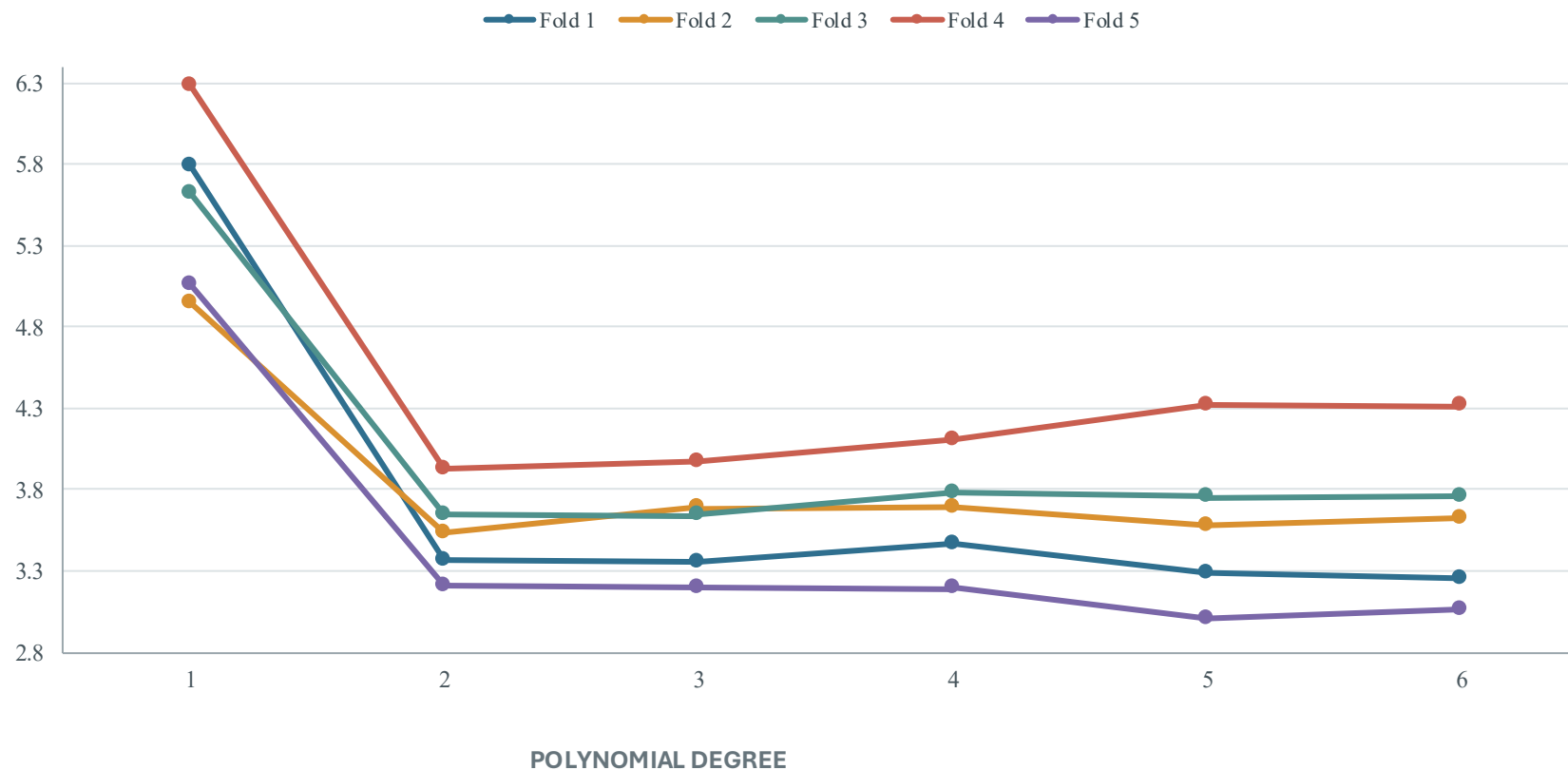
Not yet.

The ranking is real; the evidence is weak.

These models are almost indistinguishable on this particular split.

The same model performs differently across folds

VALIDATION RMSE



WHAT MOVES?

Fold composition

Fold 4 is consistently the hardest version of the experiment.

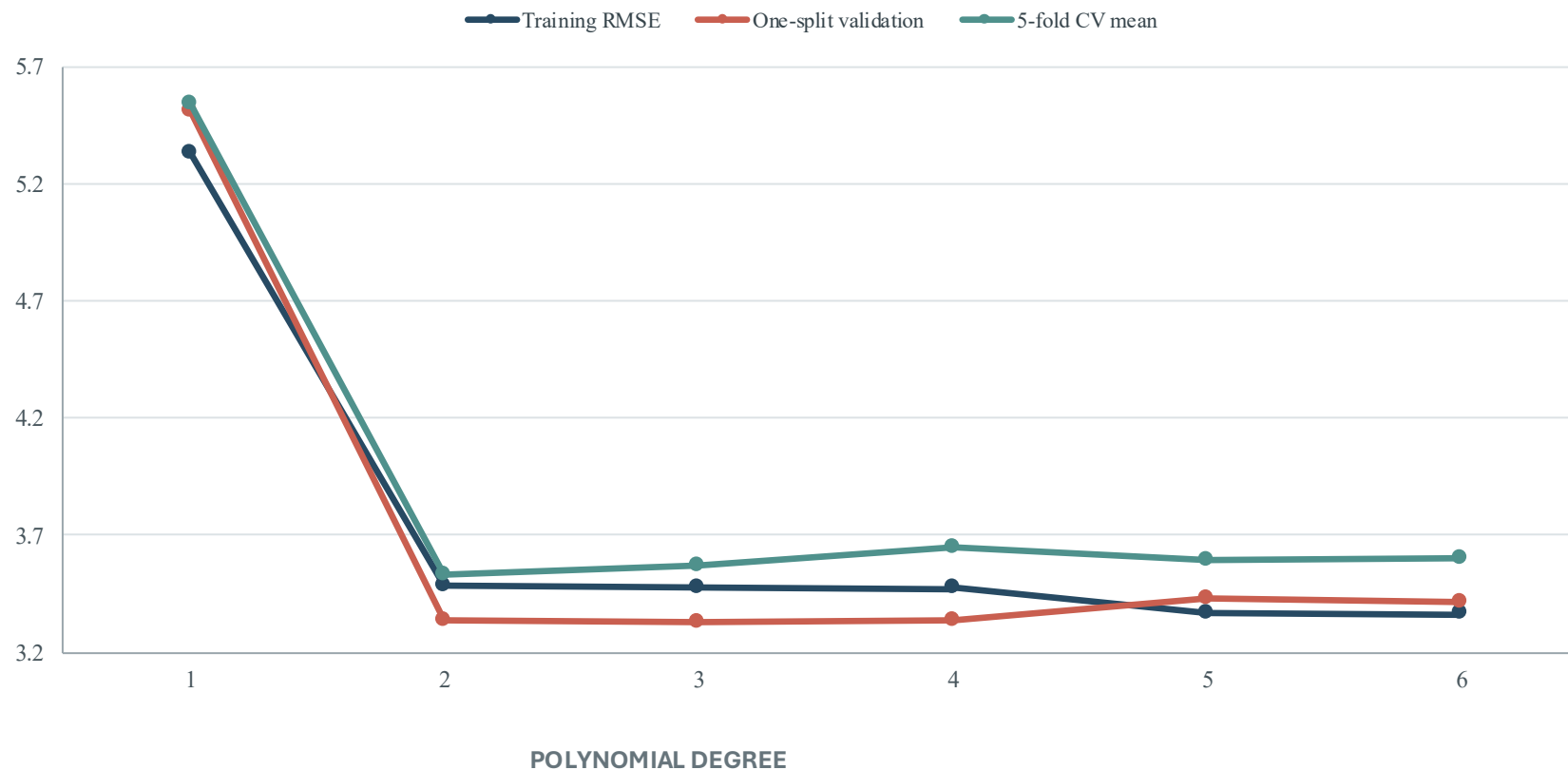
One fold can exaggerate or hide small differences between degrees.

Does the ranking survive all five versions?

Cross validation makes sensitivity visible. It does not remove it.

One split and cross validation tell different stories

RMSE

**ONE SPLIT****Degree 3**

Validation RMSE 3.333

5 FOLD CV**Degree 2**

Mean 3.535 · SD 0.276

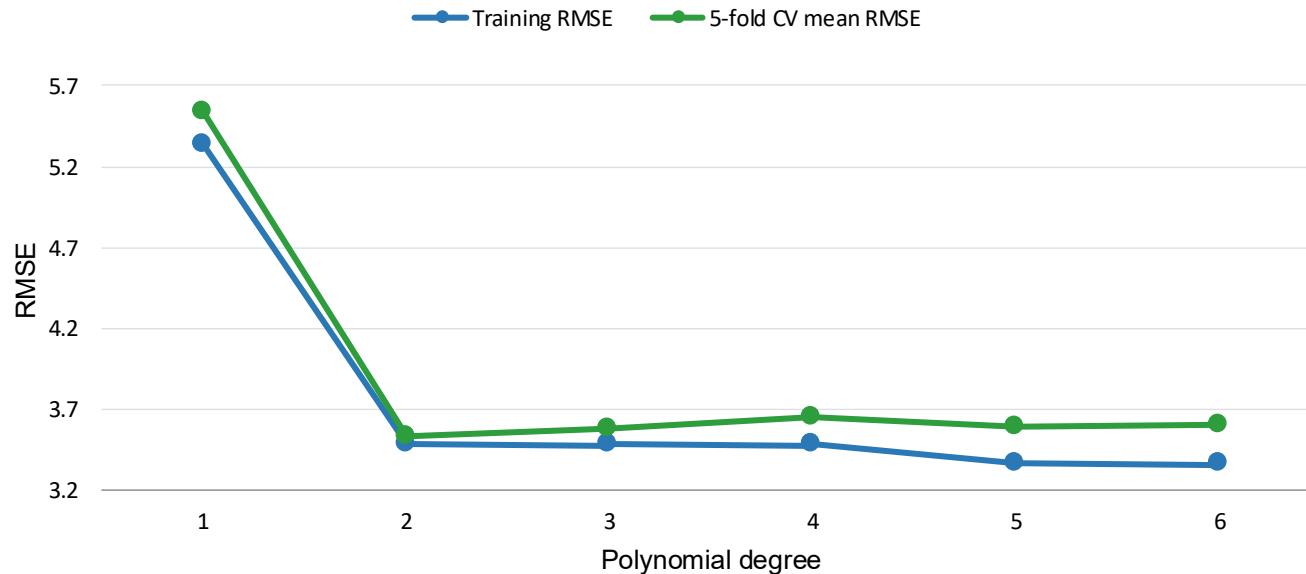
DECISION
Degree 2 is better supported

Mean performance, variability, and complexity are evidence considered together.

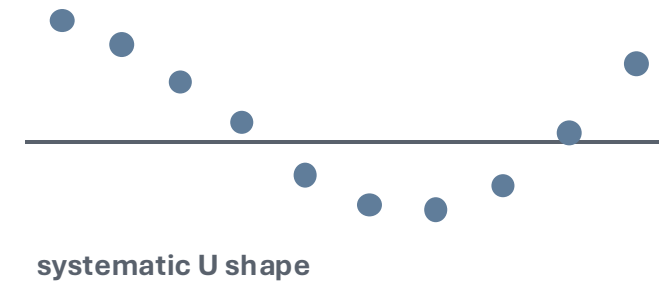
The challenge showed clear underfitting and only weak overfitting evidence

Complexity is not the diagnosis. The diagnosis comes from training and unseen data behaviour together.

Training error can keep falling while validation evidence stops improving



DEGREE 1 · RESIDUAL PATTERN



UNDERFIT

Degree 1:

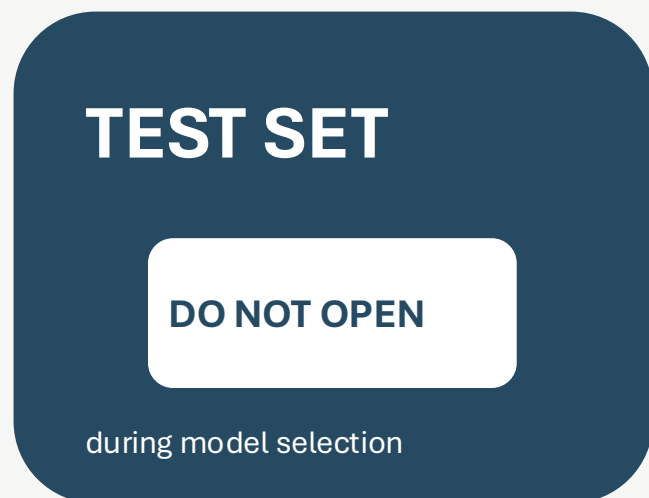
- high train error
- high CV error
- missed curvature

CAUTION

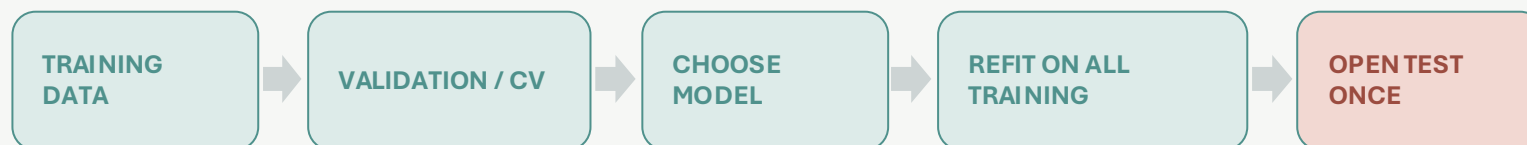
Degrees 5 and 6:
train improves,
CV does not;
spread increases

Evidence of overfitting means unseen data performance worsens as training fit improves, not simply that the degree is high.

The test set has one job



CORRECT WORKFLOW



INCORRECT PATH 1

Run CV on complete X and y after creating a test split.

Test observations enter model selection.

INCORRECT PATH 2

Evaluate degrees 1 to 6 on test; choose the smallest RMSE.

The test becomes another validation set.

VALIDATION / CV ASKS

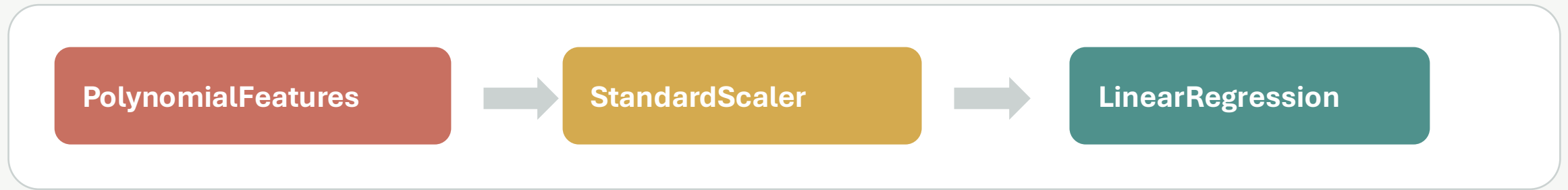
Which model should we choose?

TEST ASKS

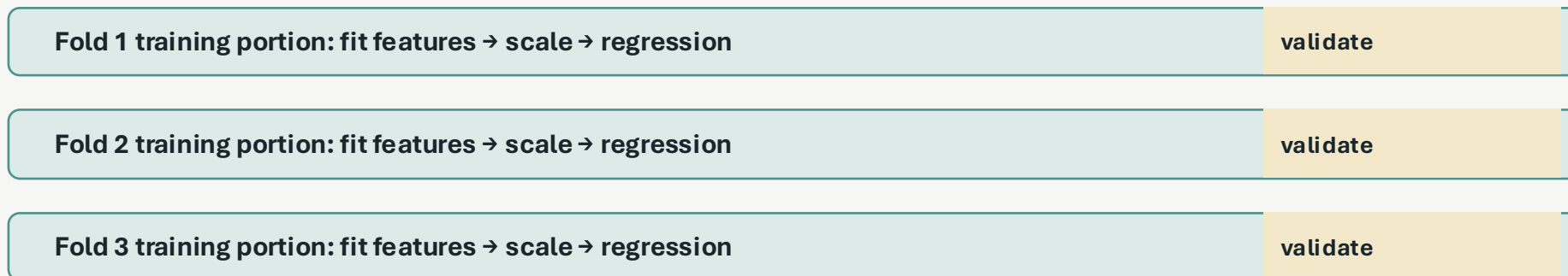
How did the chosen model generalize?

Select the workflow, not just the degree

THE MODEL IS THE WHOLE PIPELINE



FIT INSIDE EACH CV FOLD



WHY?
Any learned
step must
see only the
training
portion of
that fold.

Do not evaluate one workflow in CV and silently change it for the final test.

What I want you to do differently next time

- 1 Reserve the test set
- 2 Diagnose model inadequacy
- 3 Define candidate models
- 4 Use validation for an initial hypothesis
- 5 Treat the choice as provisional
- 6 Use CV to test stability
- 7 Compare mean, spread, and complexity
- 8 Freeze the selected workflow
- 9 Evaluate once on the untouched test
- 10 State evidence and uncertainty

Your job is not to defend your first answer. Update it when the evidence changes.

AI suggestions are hypotheses, not evidence. Check mathematical compatibility first.

Restart the runtime and run all cells top to bottom before submission.

Return to the cycling data

Day	Temperature	Rain	Wind	Decision
1	22	0	12	Outside
2	8	1	35	Inside
3	17	0	18	Outside
4	11	1	20	Inside
5	19	0	32	Outside
6	6	0	40	Inside
7	15	0	14	Outside
8	10	1	28	Inside

Suppose tomorrow is very windy. Should that change what we believe about whether the decision will be Inside or Outside?

Start from the data, not from a formula.

Start with the base rate

Day	Temperature	Rain	Wind	Decision
1	22	0	12	Outside
2	8	1	35	Inside
3	17	0	18	Outside
4	11	1	20	Inside
5	19	0	32	Outside
6	6	0	40	Inside
7	15	0	14	Outside
8	10	1	28	Inside

FOUR OF EIGHT DAYS WERE INSIDE

$$P(\text{Inside}) = 4/8 = 0.50$$

Our starting belief

Probability starts with the reference group

Day	Temp.	Rain	Wind	Decision
1	22	0	12	Outside
2	8	1	35	Inside
3	17	0	18	Outside
4	11	1	20	Inside
5	19	0	32	Outside
6	6	0	40	Inside
7	15	0	14	Outside
8	10	1	28	Inside

FOUR OF EIGHT DAYS WERE INSIDE

$P(\text{Outside}) = ?$

our starting belief · prior

DISCUSSION

Suppose tomorrow is very windy.

Should that change what we believe about Inside vs. Outside?

Create a discrete event

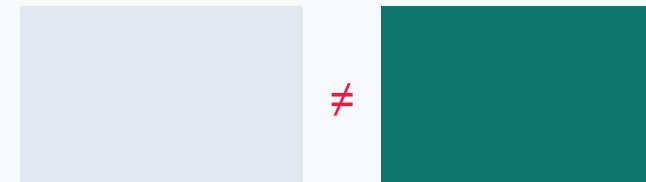
HighWind = wind > 30
km/h

Day	Temperature	Rain	Wind	Decision
1	22	0	12	Outside
2	8	1	35	Inside
3	17	0	18	Outside
4	11	1	20	Inside
5	19	0	32	Outside
6	6	0	40	Inside
7	15	0	14	Outside
8	10	1	28	Inside

$$P(\text{HighWind}) = 3/8$$

$$P(\text{HighWind} \mid \text{Inside}) = 2/4 = 0.50$$

Is this the probability we actually want?



Reversing the conditional changes the question.

Changing the condition changes the question

HighWind = wind > 30 km/h

$P(\text{HighWind} \mid \text{Inside})$

Among INSIDE days, how many had high wind?



$$2 / 4 = 0.50$$

≠

$P(\text{Inside} \mid \text{HighWind})$

Among HIGH-WIND days, how many were inside?



$$2 / 3 \approx 0.67$$

We know $P(\text{HighWind} \mid \text{Inside})$. How can we reverse the condition and find $P(\text{Inside} \mid \text{HighWind})$?

Evidence updates the probability

Among the 3 HighWind days, 2 were Inside.

$$P(\text{Inside} \mid \text{HighWind}) = 2/3 \approx 0.67$$

$$\begin{aligned} P(\text{Inside} \mid \text{HighWind}) &= P(\text{HighWind} \mid \text{Inside}) P(\text{Inside}) / P(\text{HighWind}) \\ &= (0.5)(0.5) / (3/8) = 2/3 \end{aligned}$$

Before evidence · $P(\text{Inside})$



After HighWind evidence · $P(\text{Inside} \mid \text{HighWind})$



prior · $P(\text{Inside})$

likelihood · $P(\text{HighWind} \mid \text{Inside})$

evidence · $P(\text{HighWind})$

posterior · $P(\text{Inside} \mid \text{HighWind})$

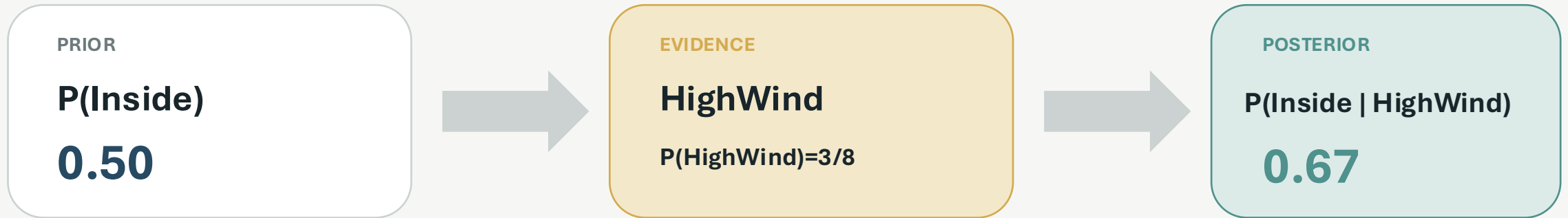
GENERAL BAYES THEOREM

$$P(h \mid D) = P(D \mid h) P(h) / P(D)$$

Why decompose it this way?

1. $P(\text{HighWind} \mid \text{Inside}) \times P(\text{Inside})$ builds the overlap: HighWind AND Inside.
2. Dividing by $P(\text{HighWind})$ asks what share of all HighWind days is in that overlap.

Evidence updates the probability



$$P(h \mid D) = P(D \mid h) \times P(h) / P(D)$$

posterior likelihood prior evidence

Bayes theorem rearranges evidence.

It does not manufacture evidence or repair weak assumptions.

NEXT: CLASSIFICATION

How do we model $P(y = 1 \mid x)$, choose a threshold, and evaluate the decision?

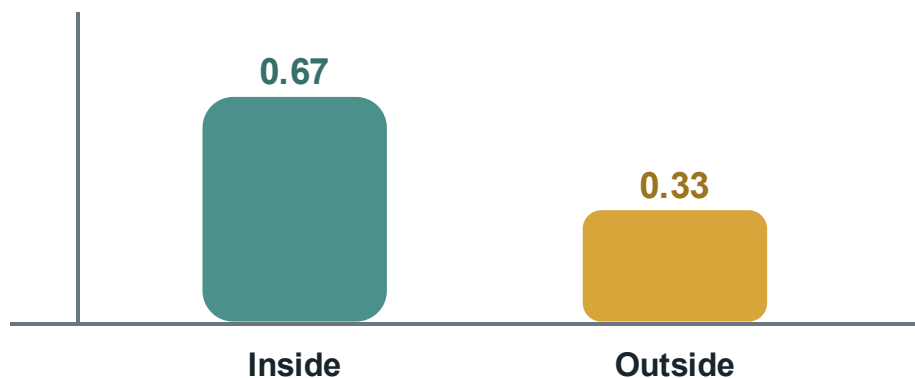
A probability is still not a decision.

Discrete probability assigns mass; continuous probability assigns density

DISCRETE VARIABLE

Countable outcomes

Tomorrow's decision is either Inside or Outside.

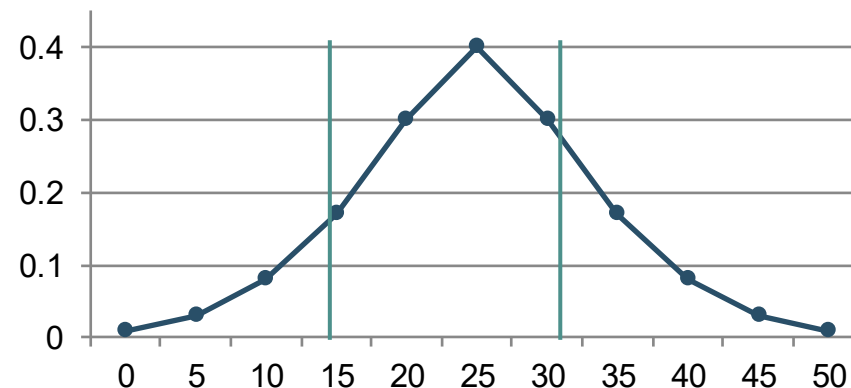


Outcomes can have positive probability. Add them: $\sum P(y) = 1$.

CONTINUOUS VARIABLE

A measurement on a continuum

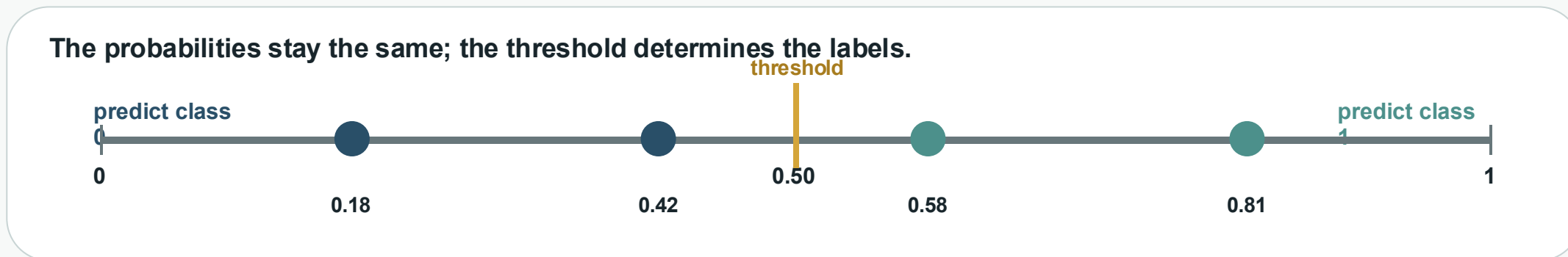
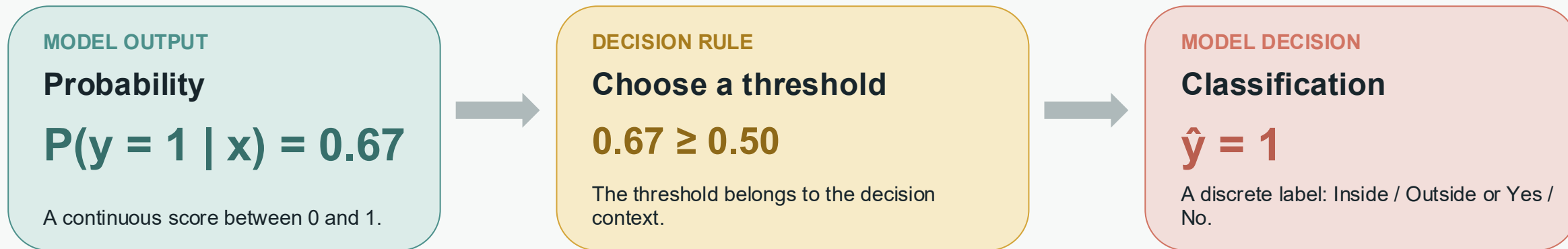
Wind speed can take any value within a range.



Height is density, not probability. One exact value has probability 0; intervals have probability.

Discrete: add probability masses. Continuous: measure area under the density.

A probability describes uncertainty; classification makes a decision



Probability asks “how uncertain?” Classification asks “which label or action?”

If missing a positive is costly, should the threshold remain 0.50?

Bayesian learning matters for practice and theory

Two reasons to study Bayesian methods

They provide practical algorithms and a disciplined way to reason about uncertainty, evidence, and model choice.

Practical learning algorithms

Methods that calculate probabilities for hypotheses, including the naive Bayes classifier, work well for many classification problems.

A framework for understanding learning

Bayesian reasoning explains why many algorithms can be viewed as selecting the hypothesis most strongly supported by the data.

The best hypothesis combines data with prior knowledge

“

Given observed training data D and a hypothesis space H , Bayesian learning selects the hypothesis with the greatest posterior probability after accounting for both the evidence and prior knowledge.

Adapted from Tom Mitchell, Machine Learning, 1997

Bayes theorem separates prior belief from new evidence

$$P(h|D) = \frac{P(D|h)P(h)}{P(D)}$$

Posterior

$P(h | D)$ is the probability of hypothesis h after observing data D .

Likelihood

$P(D | h)$ measures how well hypothesis h explains the observed data.

Prior

$P(h)$ represents belief in h before the new data is observed.

Evidence

$P(D)$ normalizes the probabilities across all hypotheses.

MAP selects the hypothesis with the largest posterior

Compare every hypothesis h in H and choose the one with the greatest posterior probability.

$$h_{\text{MAP}} = \arg \max_{h \in H} P(h|D) = \arg \max_{h \in H} P(D|h)P(h)$$

$P(D)$ can be removed because it is the same for every hypothesis and therefore cannot change which hypothesis is largest.

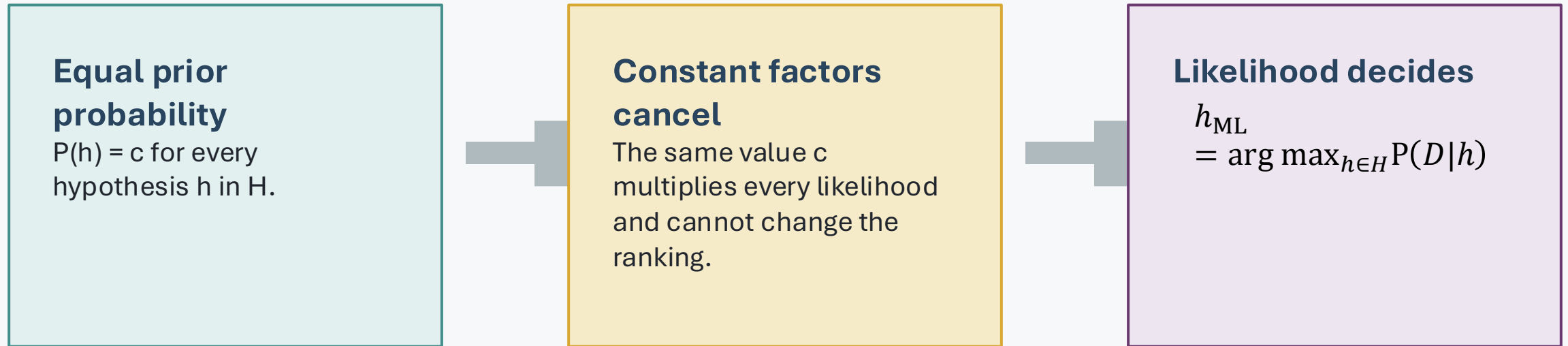
How to read **arg max**

arg identifies the input being compared.

max identifies the largest value.

The result is the hypothesis h that produces the largest posterior probability.

A uniform prior reduces MAP to maximum likelihood



Maximum likelihood chooses the hypothesis that makes the data most probable.

Least squares has a Bayesian justification under Normal noise

1. Continuous target function

The model predicts a numeric target for each training instance.

2. Independent Normal errors

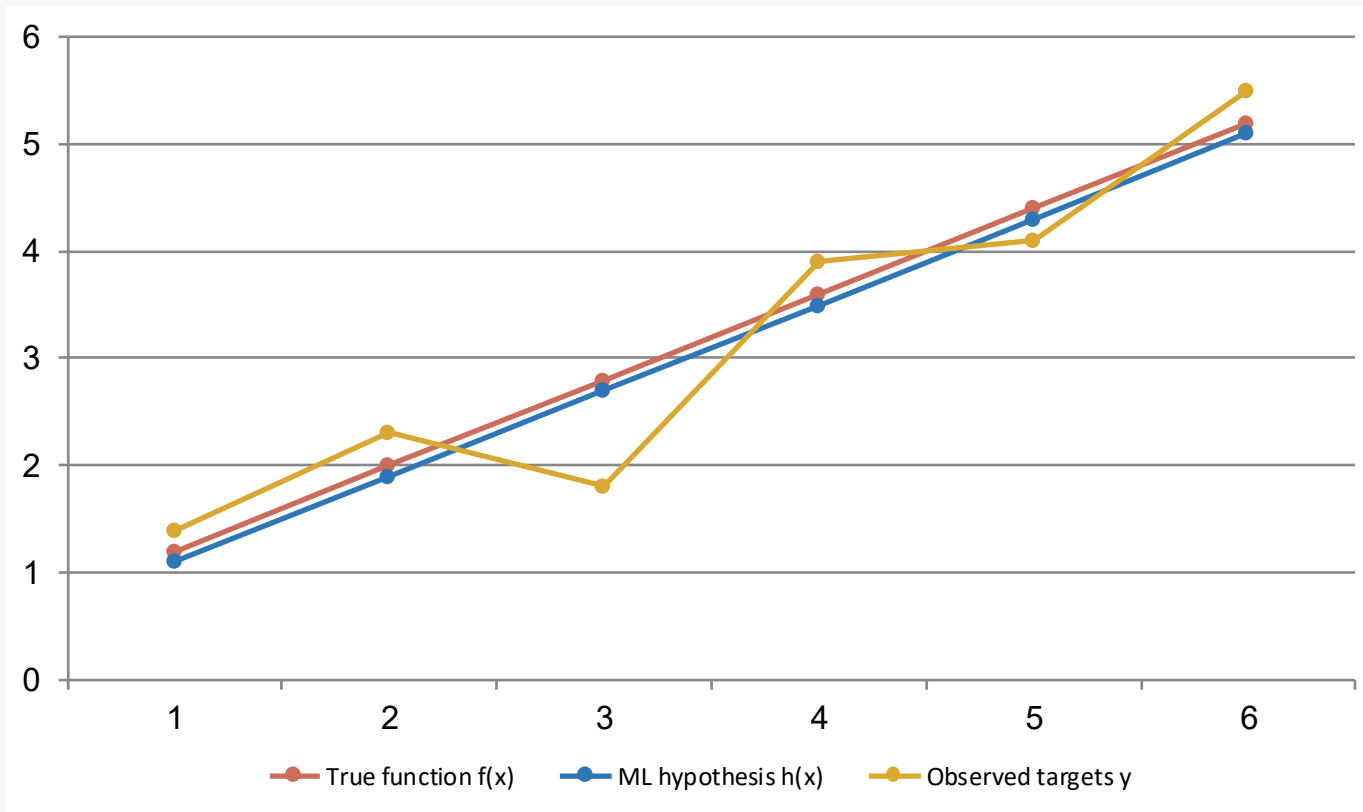
Observed targets equal the true function plus errors centered at zero.

3. Maximum likelihood result

The hypothesis that minimizes squared prediction error also maximizes the likelihood of the training data.

This connects linear regression, curve fitting, and many neural network objectives to probabilistic reasoning.

Observed targets combine a true function with random error



Illustrative training examples used to show the relationship among signal, model, and noise.

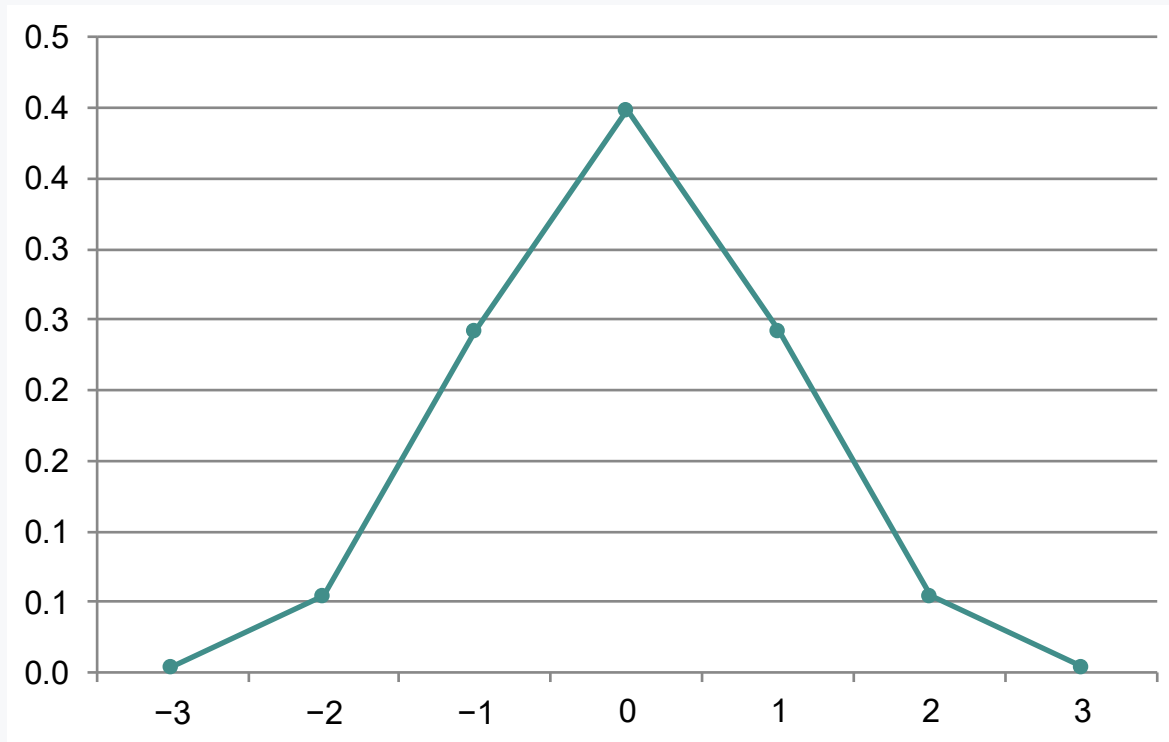
Learning a real valued function

The true function f gives the systematic target associated with each input.

Every observed target includes random error e , so the examples appear above or below the true function.

$$y^{(i)} = f(x^{(i)}) + e^{(i)}$$

Normal noise turns likelihood into a density calculation



Continuous outcomes use density

The noise follows a Normal distribution centered at zero.

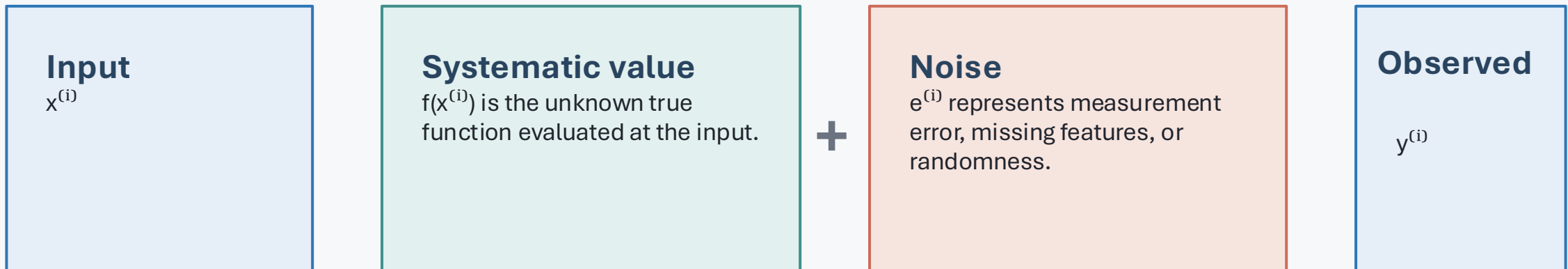
For a continuous target, $p(y | h)$ describes density near the observed value rather than probability at one exact point.

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x - \mu)^2}{2\sigma^2}\right]$$

Maximum likelihood selects the hypothesis that assigns the greatest joint density to the observed targets.

The target value separates into signal and noise

$$y^{(i)} = f(x^{(i)}) + e^{(i)}$$



Errors are assumed to be centered at zero, so they do not systematically shift the true relationship.

Independent examples make the data likelihood a product

Training inputs

$x^{(1)}, x^{(2)}, \dots, x^{(m)}$ are treated as fixed.

Observed targets

D contains $y^{(1)}, y^{(2)}, \dots, y^{(m)}$.

Conditional independence

Given h , each target contributes one density term.

$$h_{ML} = \arg \max_{h \in H} \prod_{i=1}^m p(y^{(i)} | h)$$

The full likelihood multiplies the evidence contributed by every training example.

Substituting the Normal density exposes squared error

Model assumptions

Each error $e^{(i)}$ follows a Normal distribution with mean zero and variance σ^2 .

Therefore $y^{(i)}$ is Normal with mean $f(x^{(i)})$. If h describes the target function, substitute $\mu = h(x^{(i)})$.

$$p(y^{(i)}|h) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y^{(i)} - \mu)^2}{2\sigma^2}\right]$$

$$p(y^{(i)}|h) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y^{(i)} - h(x^{(i)}))^2}{2\sigma^2}\right]$$

The likelihood becomes larger when the hypothesis prediction is closer to the observed target.

Maximizing likelihood is equivalent to minimizing squared error

1. Take the logarithm so products become sums

$$h_{\text{ML}} = \arg \max_{h \in H} \sum_{i=1}^m \left[\log C - \frac{(y^{(i)} - h(x^{(i)}))^2}{2\sigma^2} \right]$$

2. Remove constants that do not depend on h

$$= \arg \min_{h \in H} \sum_{i=1}^m \frac{(y^{(i)} - h(x^{(i)}))^2}{2\sigma^2}$$

3. Remove the positive scale factor $1 / \sigma^2$

$$h_{\text{ML}} = \arg \min_{h \in H} \sum_{i=1}^m (y^{(i)} - h(x^{(i)}))^2$$

Under the stated assumptions, the least squares hypothesis is the maximum likelihood hypothesis.

Normal noise is useful because it is tractable and widely applicable

1

Clear mathematics

The Normal density produces a likelihood that can be simplified into a familiar squared error objective.

2

Good approximation

The smooth bell shaped distribution describes many kinds of measurement and physical variation.

3

Central Limit Theorem

Sums of many independent effects often approach a Normal distribution even when the individual effects are not Normal.

Model the noise carefully, because every likelihood result depends on the assumptions behind it.